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Original research

Ankle proprioception in table tennis players: Expertise and sport-specific dual task effects

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ABSTRACT

Objectives: To compare ankle proprioception between professional adolescent table tennis players at national and regional levels and age-matched non-athletes, and, in a nominally upper-limb sport, to explore the relationships between single- and dual-task ankle proprioception, years of training and sport-specific performance.
Design: Cross-sectional observational study.
Methods: Fifty-five participants (29 professional adolescent table tennis players and 26 non-athletic peers) volunteered. Ankle proprioception was first assessed using the active movement extent discrimination apparatus (AMEDA-single) for all; yet only the players were then re-assessed while executing a secondary ball-hitting task (AMEDA-dual). The mean Area Under the Receiver Operating Characteristic Curve was calculated as the proprioceptive score, and years of training and hitting rate were recorded.
Results: National-level players had significantly better ankle proprioception as shown by higher AMEDA-single scores than the other groups (all $p < 0.05$). Ankle proprioceptive performance was significantly impaired while ball-hitting ($F_{1,28} = 58.89$, $p \le 0.001$, $\eta_p^2 = 0.69$). National-level players outperformed the regional-level significantly on the AMEDA-dual task ($F_{1,27} = 21.4$, $p \le 0.001$, $\eta_p^2 = 0.44$). Further, ankle proprioceptive performance was related to expertise, in that both AMEDA-single and AMEDA-dual proprioceptive scores were correlated with years of training and ball-hitting rate (r from 0.40 to 0.54, all $p < 0.05$).
Conclusions: Ankle proprioception is a promising measure that may be used to identify different ability levels among adolescent table tennis players. Superior ankle proprioception may arise from rigorous training and contribute to stroke accuracy. Dual-task proprioceptive assessment suggests how elite table tennis players perform differently from lower-ranked players in complex and changeable sports circumstances.

Practical implications

- Ankle proprioceptive performance is a promising measure for differentiating sports competency among adolescent table tennis players.
- Ankle proprioception was significantly associated with years of training in professional adolescent table tennis players.
- Ankle proprioceptive scores declined similarly for national- and regional-level table tennis players under the dual-task condition, suggesting that it is better to start with a higher score.

1. Introduction

In table tennis, precise and forceful ball-stroking often relies on the coordination of multiple joints, in that the upper extremity is mainly responsible for hitting the ball at the optimal timing and angle, while the lower extremities contribute to the postural stability and the transmission of information in the kinetic chain.[1] The ankle joint complex has been argued to act in the role of the only pivot, allowing the lower extremity to interact with the ground, and has the capacity to predict the movement of the other lower extremity joints.[1,2] In the somatosensory system, ankle movement control has been related to ankle proprioception, an ability to perceive the position in space and

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movement status of the ankle joint.[3,4] Previous research has mainly investigated ankle proprioception and its relationship with postural control, and sports performance in lower extremity-dominant sports.[5,6] It has not yet been determined whether ankle proprioception acuity could affect ball-stroking performance in table tennis, acknowledged as an upper extremity-dominant sport,[7] especially during real sports competitions which necessarily involves multiple-tasking.

In sports contexts, multiple-tasking is more frequent than single-tasking. A typical dual-task execution often demonstrates a central bottleneck phenomenon, whereby each task interferes with the other, due to limited central processing capacity.[8] The application of dual-task paradigms has been considered as one promising way to simulate the complexity of real sport matches, with better external validity in testing the proficiency of sport-specific techniques.[9,10] Currently, ankle proprioception has only been assessed in the single-task condition, yet this does not reflect real sports performance, and has led to the current state of limited knowledge regarding the importance of ankle proprioception in motor control. Therefore, we argue that assessing ankle proprioception during performing ball hitting may be crucial to understanding proprioceptive function in table tennis. However, traditional proprioceptive assessments seem ill-suited for the purpose of evaluating ankle proprioception during table tennis movement control.[11]

Conventionally, the methods most used in research that sets out to evaluate single joint proprioception are the assessment of threshold to detection of passive motion (TTDPM) and error in joint position reproduction (JPR). These methods involve participants being placed on isokinetic dynamometer systems so that they can either sense movements passively imposed on the ankle joint, or match previously set ankle positions, with vision and audition blocked.[12] Although these methods have been reported to be reliable,[13] both methods differ from the way in which the proprioceptive system functions during sport-related activities.[14] Consequently, it has been argued that TTDPM and JPR have low ecological validity with respect to sport settings.[14] Another method used to quantify ankle proprioceptive performance is the active movement discrimination assessment (AMEDA),[15] which allows general visual and auditory input and does not limit movement of the rest of the body. Such features enable the AMEDA to test ankle proprioception during various functional activities, resulting in improved ecological validity relative to the other two methods.[14] Although AMEDA concept variants have been successfully employed during walking and landing,[16,17] to date none have been used for testing a functional activity with a concurrent upper limb secondary task, such as hitting a ball.

It has been evident that dual-task interference can be gradually attenuated with the acquisition of the automatization, and apparent freeing-up of capacity, that comes through prolonged and intensive sport-specific training.[18] In adolescent table tennis players, it is unclear to what extent, and for how many years of sport specific training, ankle proprioceptive performance would be impaired by performing a concurrent ball-hitting task. A recent study involving a pre-school cohort suggested that sports training improved proprioceptive acuity[19]; in contrast, our previous work suggested that in adult athletes ankle proprioception is correlated with the competition level they have achieved, but not with years of training.[20] The divergence here might indicate that sport-specific training may make different contributions to proprioceptive development at different age stages. For the current study, we assume that in adolescent table tennis players, years of sport-specific training may be associated with ankle proprioception. Investigating ankle proprioception under single- and dual-task conditions, and the relationship of scores with sport-specific performance and years of training may provide critical information about the proprioceptive mechanisms underlying sport training and be relevant for talent identification.

Cumulatively, this study was designed to primarily compare ankle proprioceptive performance between professional adolescent table tennis players at national and regional levels and age-matched non-athletes, and then to explore the factors that may impact ankle proprioceptive performance. The specific hypotheses include that (1) high-ranked adolescent table tennis players would show superior ankle proprioceptive performance to lower-ranked players and age-matched nonathletic controls; (2) ankle proprioception would be correlated with ball-hitting performance, and years of training; (3) in the dual-task condition, high-ranked adolescent table tennis players would show less ankle proprioceptive performance decline than lower-ranked players.

2. Methods

2.1. Participants

A convenience sample of fifty-five adolescents volunteered, comprised of 29 professional adolescent table tennis players recruited by advertisement from the Chinese Table Tennis College, and 26 age-matched individuals without athletic experience recruited by advertising in the community. Participants were excluded if they had suffered any significant lower extremity injuries within the last three months, or had any visual, vestibular, muscular, or neurological condition that might impair balance or mobility. Thereafter, all table tennis players were classified by their given athletic levels, i.e., national-(L1) and regional-level group (L2), while the nonathletic aged-matched cohort was split into groups by their age median, i.e., an older group (C1) and a younger group (C2). Demographic information was collected from eligible participants, including their age, gender, competition level and years of sport-specific training (YOT) (Table 1). Dominant side was determined by use of the Chinese-version of the Edinburgh handedness inventory for the hand (EHI) and the Waterloo Footedness Questionnaire-revised for the foot (WFQ).[21]

2.2. Equipment

A purpose-built Active Movement Extent Discrimination Apparatus (AMEDA) was employed to provide ankle movement discrimination scores that represent participants' proprioceptive sensitivity when discriminating between four different active ankle inversion depths, in full weight-bearing stance. The reliability of the scores (Intraclass Correlation Coefficient) generated by the ankle AMEDA tests has been determined as 0.89.[22] The AMEDA device consists of a fixed wooden platform housing a swinging square wooden plate that rotates around an axle aligned to the long axis of the foot being tested. During testing, participants were instructed to look straight ahead and stood on the device in an even weight-bearing stance, with their supporting foot on the fixed platform and their testing foot centered over the axle of the swinging plate, in preparation to move the plate downward in an ankle inversion movement. A set of physical stops under the plate were used to give four inversion depths, from 10° (Position1) to 16° (Position 4), with an increment of 2° per position.

2.3. Procedure

The protocol for each testing occasion consisted of a familiarization session and a data collection session. Firstly, participants experienced the four ankle inversion positions three times in order, from Position 1 to Position 4, with 12 trials in total. After familiarization, participants then undertook a 40-trial test, where each of the four positions was randomly presented 10 times. In each trial, participants were asked to orally report their perceived position using number (1, 2, 3, or 4), without any feedback being given.

In the formal testing session, the participants were instructed to stand on the AMEDA device in front of the table, with their dominant ankle being tested. L1 and L2 performed AMEDA tests under single and dual-task conditions, with the order from a random sequence to counteract any potential learning or fatigue effects. Participants were given a 5-min break between tests. By contrast to the athletic cohorts, C1 and C2 were only tested under single-task condition, given that

Table 1
Demographic information and main outcome measures (mean ± SD).

Variables	L1	L2	C1	C2
Participants, n.	15	14	13	13
(Gender)	(8F:7M)	(2F:12M)	(4F:9M)	(8F:5M)
Age, yrs.	14.3 ± 0.6[a]	11.1 ± 1.5	14.2 ± 2.0[c]	9.9 ± 2.5
(Min–max)	(12, 17)	(9, 14)	(13, 15)	(8, 13)
YOT, yrs.	7.5 ± 1.9[a]	4.1 ± 1.0		
(Min, max)	(4, 11)	(4.5, 6)		
AMEDA-single	0.83 ± 0.07[a,b]	0.74 ± 0.07[b]	0.74 ± 0.04	0.73 ± 0.05
(Min, max)	(0.71, 0.90)	(0.69, 0.81)	(0.65, 0.79)	(0.67, 0.80)
AMEDA-dual	0.70 ± 0.08 (0.61, 0.83)	0.61 ± 0.06		
(Min, max)		(0.5, 0.7)		
HR, %	37.00 ± 14.52[a]	14.28 ± 11.62		
(Min, max)	(12.50, 52.50)	(5.00, 38.00)		

HR: Hitting rate = Times of hitting the target zone / Total times ∗ 100%.
L1: national level, L2: regional level; C1: older age-matched comparator. C2: younger age-matched comparator.
[a] Statistically different between L1 and the other group(s) (all $p < 0.0001$ or 0.05).
[b] Statistically different between AMEDA and AMEDA-dual ($p < 0.05$).
[c] Statistically different between C1 and C2 ($p < 0.0001$).

they had little table tennis experience. Under the single-task condition, participants were instructed to do the AMEDA-single test in front of a table-tennis table. Wooden blocks were placed under the tennis table legs to address the height difference created by the AMEDA device. For the dual-task condition, participants completed the AMEDA test and simultaneously hit table tennis balls with a standard forehand attack technique (AMEDA-dual). There were 40 balls to hit in total, released by a table tennis robot (Y&T® V-988, Guangdong, China) located 1 m away from the opposite edge of the table and set to release topspin balls, with an initial speed of 45 in./s.

2.4. Data analysis

To obtain proprioceptive movement discrimination scores, each participant's raw data were first cast into a 4 × 4 matrix representing the frequency with which each response was made to each stimulus. Non-parametric signal detection analysis was used to produce three pair-wise Receiver operating characteristic curves (ROC). The mean area under the curve (AUC) was calculated as the ankle proprioceptive score, providing an unbiased estimate of the ability of an individual to discriminate between adjacent pairs of stimuli.[19] AUC values range from 0.5, equivalent to chance responding, up to 1.0, representing perfect discrimination. To evaluate stroking performance, there were two target zones (30 cm ∗ 30 cm) marked on the distal table corners (Fig. 1). Players were instructed to hit the ball into the diagonal target zone, i.e., right-handed players hit the balls diagonally towards the left corner target zone and vice versa. The hitting rate (HR) was calculated by the actual times of hitting the target zone being divided by the total times, which was used to represent a player's ability to stroke the ball into a desired area of the table.

Two-tailed independent-sample *t*-tests were utilized to compare the AMEDA scores between adolescent table tennis players and the nonathletic age-matched cohort. For analysis of ankle proprioception scores under varying test conditions in table tennis groups (L1 and L2), repeated measures analysis of variance (RM-ANOVA) was performed to compare the differences between groups and between the 2 AMEDA testing conditions. Partial Eta Squared (η_p^2) was calculated as the effect size index. In addition, the relationship between ankle AMEDA movement discrimination scores, YOT and HR was further explored. Pearson's correlation analysis was used for continuous data, and the Spearman method for non-continuous data. All statistical analyses were performed using SPSS 25(SPSS Inc., Chicago, IL, USA). The significance level was set at $p < 0.05$.

3. Results

Demographic information for the fifty-five recruited adolescent table tennis players and the age-matched non-athletes is presented in Table 1. Only one table tennis player was left-dominant, while the other 28 players and all non-athletes were right-dominant. L1 and C1 were significantly older than L2 and C2 (both $p < 0.001$), but there was no significant age difference between L1 and C1, or L2 and C2 (both $p > 0.05$), allowing age-matched comparisons regarding ankle discrimination ability between adolescent table tennis players and non-athletic cohort.

On the AMEDA scores from the single-task condition, L1 performed significantly better than L2 ($t_{27} = 3.48$, $p < 0.05$, 95%CI (0.03, 0.13)), C1 ($t_{26} = 3.86$, $p \le 0.001$, 95%CI (0.04, 0.12)) and C2 ($t = 3.35$, $p < 0.05$, 95%CI (0.05, 0.14)) (Table 1). RM-ANOVA showed a significant testing condition main effect ($F_{1,28} = 58.89$, $p \le 0.001$, $\eta_p^2 = 0.69$), and a competition level main effect ($F_{1,27} = 21.4$, $p \le 0.001$, $\eta_p^2 = 0.44$), with no interaction effect ($p = 0.907$). These results suggest that the AMEDA-dual performance was significantly worse than AMEDA-single performance, that high-level table tennis players (L1) performed significantly better under the two AMEDA testing conditions, and that these affects were independent of each other (Fig. 2). In general, both high and low-level table tennis players showed a dual-task cost of approximately 15% compared to their single-task AMEDA scores.

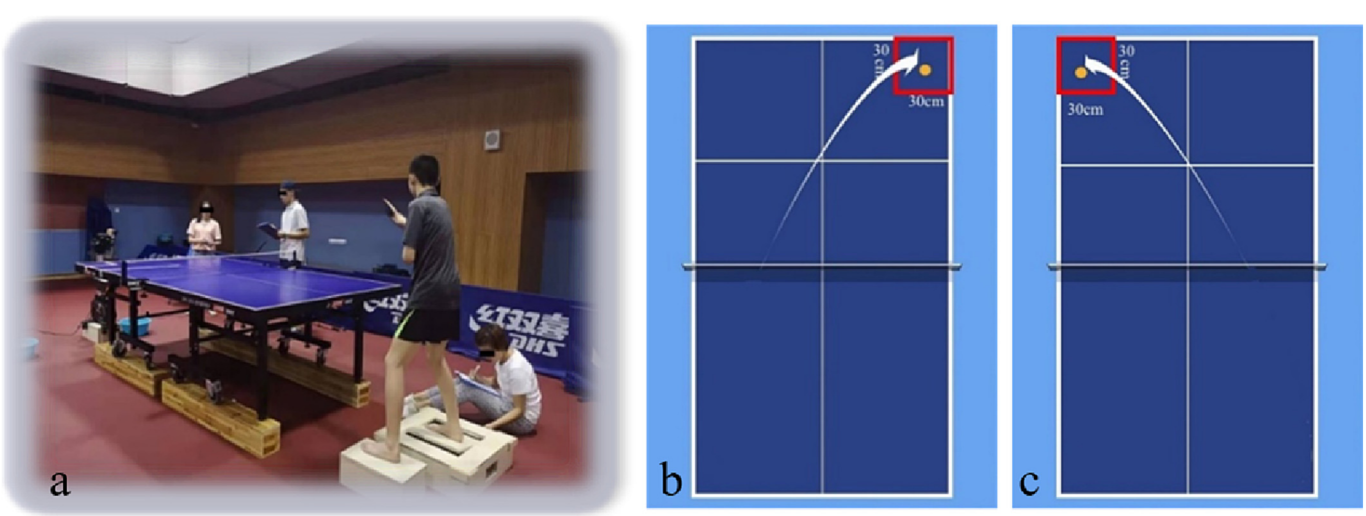

Fig. 1. Ankle proprioception testing setup under the dual-task condition (AMEDA-dual). Testing scene (a); left-handed players strike zone (b); right-handed players strike zone (c).

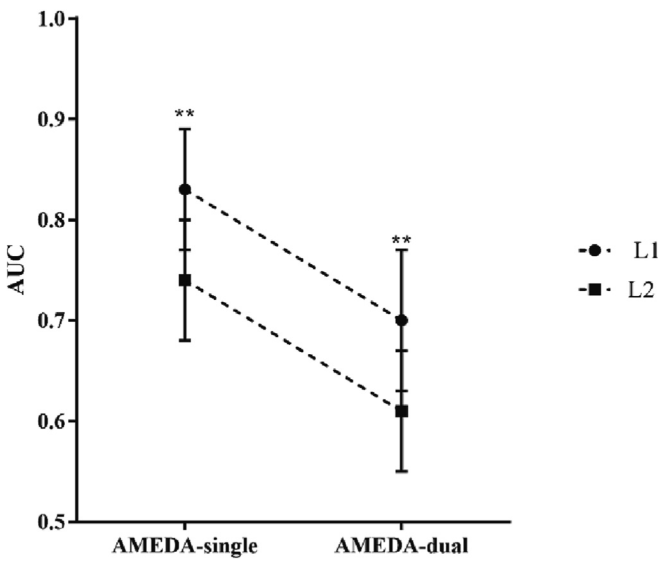

Fig. 2. Comparisons of mean ankle movement discrimination scores with standard deviations between different testing conditions for adolescent table tennis players.
L1: national level, L2: regional level.

With respect to correlations between ankle proprioception, HR and YOT, Pearson's correlation analysis showed that both AMEDA-single and AMEDA-dual scores were significantly and positively correlated with HR (r ranged from 0.40 to 0.54, all $p < 0.05$), indicating that better performers had better ankle proprioception. Non-parametric correlation analysis showed that YOT was significantly correlated with both AMEDA-single (Spearman's rho $= 0.45$, $p < 0.05$) and AMEDA-dual scores, as well as HR (Spearman's rho $= 0.67$, $p < 0.05$), suggesting a training effect underlying the sport-specific performance of higher-level players.

4. Discussion

This was the first study designed to primarily investigate ankle proprioceptive performance during single- and dual-task conditions among adolescent table tennis players and then to examine the relationship between ankle proprioception and sport-specific training. The current results suggest that national level table tennis players demonstrate superior ankle proprioception when compared with age-matched non-athletes and regional level players. Further, there were significant positive correlations between single- and dual-task ankle proprioception scores, HR, and YOT measures among adolescent table tennis players. Relative to regional level athletes, national level players performed significantly better on the AMEDA task, single or dual.

Consistent with previous studies,[23,24] results from this study show that ankle proprioceptive assessment can be used to distinguish high-ranked adolescent table tennis players from their lower-ranked and nonathletic counterparts. Results here also support the previously observed connections between ankle proprioception and lower extremity motor control.[4] Despite table tennis being an apparently upper extremity dominant sport, it requires sophisticated lower extremity motor control to maintain dynamic balance and enable mastery of co-ordination in sport performance.[25] Specifically, findings demonstrated that national-level athletes performed significantly better on the ankle proprioception task, and that there was a significant positive correlation between ankle discrimination sensitivity and table tennis stroking performance. Superior proprioceptive ability in elite athletes has been attributed to prolonged sport-specific training.[24,26] Our results support this notion, as we also found a significant correlation between ankle proprioception scores and YOT. However, regional level players with approximately 4.1 YOT did not perform better than their age-matched nonathletic counterparts.

Rigorous sport-specific training (longer training sessions, higher training intensity, and fiercer competitions) associated with elite performance, in addition to sport-specific training,[27] may account for the differences noted between the AMEDA scores in national- and regional-level players. Fundamental differences in training may contribute to a higher level of proprioceptive performance. Another explanation is that ankle proprioception ability may be a biological attribute.[14] Han et al.[14,20] found higher-level adult athletes to have better ankle proprioception, however no association between ankle proprioception and YOT was found. Consequently, they argued that ankle proprioception may be more genetically than environmentally-determined.[14] In this study, mean proprioceptive scores for the regional level players were similar to those of their age-matched counterparts, however the standard deviation associated with the mean for the players was much larger. This suggests that some players competing at the regional level already had relatively high ankle proprioceptive ability. Therefore, based on selection of the fittest, it is possible that those with poor ankle proprioceptive ability drop out because they have difficulties in coping with rigorous training, in further improving their performance, or from having injuries acquired in their efforts to succeed. Longitudinal studies are needed to investigate to what extent ankle proprioceptive ability can be improved by rigorous training, and how this improvement is associated with table tennis performance.

Furthermore, ankle proprioception of adolescent table tennis players was assessed under dual-task conditions. Compared with single-task assessment, the dual-task condition tends to be more challenging, but it better reflects the way that the central nervous system must work to process proprioceptive information in complex daily tasks and sports tasks.[28,29] In previous research,[4,20] Han et al. found that high-ranked athletes had superior ankle proprioception and argued that superior ankle proprioceptive ability enables athletes to perform a secondary sport-related task, such as controlling a soccer ball while allocating capacity to processing information about team members and thereby determining the best timing to pass the ball. In the current study, dual-task execution involved controlling ankle movement while performing table tennis ball stroking. Our results suggested that under dual-task assessment, HR was moderately associated with both AMEDA-single and AMEDA-dual performance. These findings are consistent with a previous study that involved a mixed group of athletes aged from 18 to 25, where the authors found competition level achieved, from regional to Olympic level could be predicted by ankle proprioceptive performance.[20] Therefore, the current findings replicate and extend the notion that ankle proprioception, as measured under single- and dual-task conditions, may be fundamental for accurate stroke performance in adolescent table tennis players. Further, YOT were moderately correlated with HR under the dual-task condition, and strongly with ankle discrimination ability when compared with that under the single-task condition. This points to the importance of sports training on dual-task execution incorporating ankle proprioception, and the underlying explanation may be that intensive and prolonged sports-specific training can facilitate the automation of sports skills, thereby enabling athletes to allocate more central processing to novel or challenging concurrent tasks.[25]

In concordance with the central bottleneck model,[8] the current data also suggested that in adolescent table tennis players, the overall ankle proprioception scores under the dual-task condition declined by approximately 15% compared to scores obtained in the single-task condition. Perhaps by involving more complicated tasks to be performed concurrently, the dual-task cost is slightly higher than the reported by Schafer et al.,[9,10] with 0 to 10% for expert players in response to hitting a table tennis stroke while counting backwards. When comparing the two groups, against our hypothesis, the high-ranked players, although having a higher hitting accuracy demonstrated similar dual-task cost on ankle proprioception scores, which may suggest their task prioritization in favor of the sport-specific task.[30] Nevertheless, high-ranked players demonstrated consistently superior ankle proprioceptive ability either in a single- or dual-task condition. Cumulatively, these findings indicate that having superior ankle proprioception may contribute to maintaining high-level competitive status for adolescent table tennis players, especially when processing demands are increased in complex and variable sports circumstances.

There are some limitations in this study. One is that the design does not fully replicate table tennis playing conditions and stroke repertoires. Actual conditions of table tennis competitions can be more complex than our current setup. In addition to the frequent footwork switching in games, table tennis ball parameters, such as speed, trajectory, and spin, could be dynamically changed. In this study, participants were instructed to perform the tests in static standing, and the ball speed was controlled by using a robot server, therefore outcomes may be different from the actual dual-task performance required in table tennis games.

5. Conclusion

Ankle proprioception is a promising measure that may be used to identify different ability levels among adolescent table tennis players. Superior ankle proprioception may arise from rigorous training and contribute to stroke accuracy. Dual-task proprioceptive assessment suggests how elite table tennis players perform differently from lower-ranked players in complex and changeable sports circumstances.

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Confirmation of ethical compliance

This study was approved by the Shanghai University of Sport Ethics Committee for Human Research (approval number: 102772020RT008). Prior to testing sessions, written informed consent was obtained from all participants, as well as their guardians.

CRediT authorship contribution statement

X. Shi and J. Han conceived and designed research; X. Shi and Z. Cao performed data collections. X. Shi analyzed data; X. Shi, and J. Han interpreted results of the experiments; X. Shi drafted the manuscript, X. Shi, C. Ganderton, O. Tirosh, R. Adams, DE. Ansary and J. Han edited and revised the manuscript, X. Shi, DE. Ansary, R. Adams and J. Han approved the final version of the manuscript.

Declaration of interest statement

The authors declare no conflict of interest in this study.

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